A Comparison of Covariance Matrix Shrinkage Methods for Portfolio Optimization under a Low Asset-to-Observation Ratio

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Удиртгал

This study compares alternative covariance matrix shrinkage estimators in the context of portfolio selection. The performance of the estimators is evaluated using the Global Minimum Variance Portfolio (GMVP).

Keywords: Covariance matrix, Shrinkage estimation, Portfolio selection, Global minimum variance portfolio, Ledoit–Wolf.

1 Introduction

Portfolio optimization requires reliable estimates of the covariance matrix of asset returns.

The traditional sample covariance matrix may perform poorly, particularly when the number of assets is large relative to the number of observations.

Shrinkage estimators provide an alternative approach by combining the sample covariance matrix with a structured target matrix.

The main objective of this study is to compare different covariance matrix shrinkage methods in the context of minimum-variance portfolio selection.

2 Literature Review

Covariance matrix estimation has received considerable attention in portfolio theory.

The shrinkage approach was introduced into portfolio applications by Ledoit and Wolf (2004).

More recent studies have considered both linear and nonlinear shrinkage estimators.

3 Methodology

3.1 Sample Covariance Matrix

Let

$$R_t = \begin{pmatrix} R_{1t} \\ R_{2t} \\ \vdots \\ R_{Nt} \end{pmatrix}$$

denote the vector of asset returns.

The sample covariance matrix is

$$S = \frac{1}{T} \sum_{t=1}^{T} (R_t - \bar{R})(R_t - \bar{R})'. \tag{1}$$

3.2 Linear Shrinkage Estimation

The general linear shrinkage estimator is defined as

$$\widehat{\Sigma} = \delta F + (1 - \delta) S, \tag{2}$$

where

- S is the sample covariance matrix,
- F is the shrinkage target matrix,
- $\delta \in [0, 1]$ is the shrinkage intensity.

3.3 Shrinkage Targets

Several target matrices are considered in this study.

3.3.1 Constant Variance Target

3.3.2 Diagonal Target

3.3.3 Constant Correlation Target

3.3.4 Single-Factor Target

3.4 Nonlinear Shrinkage

The nonlinear shrinkage estimator can be expressed as

$$\widehat{\Sigma}_{NLS} = U\widehat{D}U', \tag{3}$$

where U contains the eigenvectors of the sample covariance matrix and

$$\widehat{D} = \mathrm{diag}(\widehat{d}_1, \ldots, \widehat{d}_N)$$

contains the shrunk eigenvalues.

4 Portfolio Optimization

This study considers the Global Minimum Variance Portfolio (GMVP).

The optimization problem is

$$\min_{w} \quad w'\widehat{\Sigma}w \tag{4}$$

subject to

$$\mathbf{1}'w = 1. \tag{5}$$

The analytical solution is

$$w_{\mathrm{GMVP}} = \frac{\widehat{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}'\widehat{\Sigma}^{-1}\mathbf{1}}. \tag{6}$$

5 Data

This section describes the dataset used in the empirical analysis.

Хүснэгт 1: Descriptive Statistics

Asset	Mean	Std. Dev.	Minimum	Maximum
Asset 1	0.00	0.00	0.00	0.00
Asset 2	0.00	0.00	0.00	0.00
Asset 3	0.00	0.00	0.00	0.00

6 Empirical Results

6.1 Covariance Matrix Estimation

The estimated covariance matrices are compared in terms of their statistical properties.

6.2 Portfolio Weights

The GMVP weights obtained from different covariance estimators are compared.

6.3 Portfolio Performance

Portfolio performance is evaluated using volatility, variance, Sharpe ratio, and portfolio turnover.

Хүснэгт 2: GMVP Performance Comparison

Estimator	Return	Volatility	Sharpe Ratio	Turnover
Sample Covariance				
Constant Variance				
Diagonal				
Constant Correlation				
Single Factor				
Nonlinear Shrinkage				

7 Discussion

This section discusses the empirical differences between the covariance matrix estimators and their implications for portfolio selection.

8 Conclusion

This study compares several covariance matrix shrinkage methods for constructing global minimum variance portfolios.

The empirical results demonstrate how covariance estimation affects portfolio risk, asset allocation, and portfolio stability.

Ашигласан ном

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